UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
**United States Patent and Trademark Office**
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

| APPLICATION NO. | FILING DATE | FIRST NAMED INVENTOR | ATTORNEY DOCKET NO. | CONFIRMATION NO. |
|---|---|---|---|---|
| 17/590,044 | 02/01/2022 | Teemu KORPIMÄKI | 9888.34 | 8566 |

15308 7590 12/20/2023

MYERS BIGEL, P.A.
4130 Parklake Avenue
Suite 250
Raleigh, NC 27612

| EXAMINER | |
|---|---|
| HAROUAKA, RAMDANE | |

| ART UNIT | PAPER NUMBER |
|---|---|
| 1677 | |

| NOTIFICATION DATE | DELIVERY MODE |
|---|---|
| 12/20/2023 | ELECTRONIC |

**Please find below and/or attached an Office communication concerning this application or proceeding.**

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspto@myersbigel.com

| Office Action Summary | Application No. 17/590,044 | Applicant(s) KORPIMÄKI et al. | |
|---|---|---|---|
| | Examiner Ramdane Harouaka | Art Unit 1677 | AIA (FITF) Status Yes |

*-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --*

**Period for Reply**

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

**Status**

1)☑ Responsive to communication(s) filed on 11/13/2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

**Disposition of Claims***

5) ☑ Claim(s) 1-23 is/are pending in the application.

5a) Of the above claim(s) 14-23 is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-13 is/are rejected.

8) ☑ Claim(s) 5 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

**Application Papers**

10)☑ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 02/01/2022 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

**Priority under 35 U.S.C. § 119**

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

**Certified copies:**

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

**Attachment(s)**

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 11/18/2022 and 09/06/2023.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

## DETAILED ACTION

### *Notice of Pre-AIA or AIA Status*

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

### *Election/Restrictions*

Applicant's election without traverse of Group I, claims 1-13, in the reply filed on 11/13/2023 is acknowledged. Claims 14-23 are withdrawn from further consideration pursuant to 37 CFR 1.142(b) as being drawn to a nonelected invention, there being no allowable generic or linking claim.

### *Priority*

This application, Serial No. 17/590,044 was filed on 02/01/2022 and claims benefit to US Provisional Application Serial No. 63/144,215 filed on 02/01/2021.

### *Status of the Claims*

Acknowledgement is made of amendments to the claims dated 11/13/2023. Claims 1-23 are pending. Claims 14-23 are withdrawn as non-elected claims. Claims 1-13 are examined herein.

### *Information Disclosure Statement*

The information disclosure statements filed on 11/18/2022 and 09/06/2023 are in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statements have been considered by the examiner.

### *Specification*

The abstract of the disclosure is objected to because it exceeds 150 words in length. Correction is required. See MPEP § 608.01(b).

The disclosure is objected to because of the following informalities:

Para. 37 and Para. 49 reference "a UNCP" and "UNCPs" respectively. The term UNCP is not defined in the disclosure, and appears to be a misspelling of the term UCNP.

Appropriate correction is required.

The use of the term UPCON (See specification para. 33), which is a trade name or a mark used in commerce, has been noted in this application. The term should be accompanied by the generic terminology; furthermore the term should be capitalized wherever it appears or, where

appropriate, include a proper symbol indicating use in commerce such as ™, SM , or ® following the term.

Although the use of trade names and marks used in commerce (i.e., trademarks, service marks, certification marks, and collective marks) are permissible in patent applications, the proprietary nature of the marks should be respected and every effort made to prevent their use in any manner which might adversely affect their validity as commercial marks.

### *Claim Objections*

Claim 5 is objected to because of the following informalities:

Claim 5 recites the limitation "wherein detecting a UNCP label of the UNCP labeled conjugate" in lines 1-2. The claim is objected to because of apparent typographical errors, namely that "UNCP" appears to be a misspelling of the term "UCNP". Note that for the purpose of continued examination, the references to UNCP have been interpreted as referring to UCNP.

Appropriate correction is required.

### *Claim Rejections - 35 USC § 103*

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

Claims 1-6, 8, and 11 are rejected under 35 U.S.C. 103 as being unpatentable over **Magliulo et al** (Magliulo, Maria, et al. "A rapid multiplexed chemiluminescent immunoassay for the detection of Escherichia coli O157: H7, Yersinia enterocolitica, Salmonella typhimurium, and Listeria monocytogenes pathogen bacteria." Journal of agricultural and food chemistry 55.13 (2007): 4933-4939.) as cited on the IDS dated 11/18/2022, in view of **He et al** (He, Hao, et al. "Quantitative lateral flow strip sensor using highly doped upconversion nanoparticles." Analytical chemistry 90.21 (2018): with Supplemental Information, 17 pages).

**Magliulo et al** teach throughout the document a method for detecting target analytes from a sample using spatial multiplexed assays on a microplate comprising main wells with four

subwells each that are conjugated with specific binding reagents (Fig. 1, Abstract, and pp. 4934, col. 1, para. 4: In this work, we report the development of a simple, multiplexed sandwich chemiluminescent enzyme immunoassay (CL-EIA) for the simultaneous detection of four of the major foodborne pathogens: E. coli O157:H7, Y. enterocolitica, S. typhimurium, and L. monocytogenes.).

Regarding claims 1 and 11, Magliulo et al further teach:

providing a microplate comprising: a plurality of wells arranged in a two-dimensional array, each of the wells comprising: a bottom surface, sidewalls extending from the bottom surface to form an open top; at least two subwells in the bottom surface, the at least two subwells having sidewalls that extend below the bottom surface of the well (Fig. 1 shows a microplate provided for analyte detection comprising 96 wells arranged in a two-dimensional array, with each well comprising four subwells; and pp. 4934, col. 2, para. 6, lines 2-9: To achieve the simultaneous detection of the four pathogens in the same sample, a new polystyrene 96 x four well microtiter plate format was developed (Figure 1A). The prototype microtiter plate was fabricated in our lab by modifying a commercial 384 well polystyrene microtiter plate (Labsystems). The 96 main wells were obtained by drilling the plate with a cylindrical milling cutter (Ø = 6 mm) to obtain a main well among each of the four wells. While the depth of the wells of the microtiter plate was 8 mm, the drilling depth was only 7 mm; thus, four small independent subwells remained in the bottom; Therefore, each of the 96 main wells comprises a bottom surface, sidewalls that extend 7 mm above the bottom to form an open top, and four subwells with sidewalls that extend 1 mm below the bottom), wherein each of the at least two subwells comprises a capture binding agent that is configured to bind to a target analyte, if

present, in a sample (Abstract lines 3-6: To achieve the multiplexed detection of the four pathogens, a new polystyrene 96 well microtiter plate format has been designed, in which each main well contains four subwells in the bottom. The monoclonal antibodies specific for each bacteria were separately immobilized in each subwell.);

adding a sample to the well such that the sample fluidically contacts each of the at least two subwells such that a target analyte, if present, binds to the capture binding agent in one or more of the at least two subwells, wherein **a peroxidase-labeled conjugate** binds with the target analyte, if present; and detecting a label of the **peroxidase-labeled conjugate** at one or more of the at least two subwells to thereby determine whether the target analyte is present in the sample (Abstract lines 6-11: When the samples were added to the main wells, the bacteria able to specifically bind to the corresponding monoclonal antibody were captured in one of the four subwells. Subsequently, a mixture of peroxidase-labeled polyclonal antibodies against the four bacteria was added and the peroxidase activity of the bound polyclonal labeled antibodies in each well was measured by an enhanced luminol-based chemiluminescent cocktail using a low-light charge-coupled imaging device.).

Magliulo et al fail to teach:

adding the **upconverting nanoparticle (UCNP) labeled conjugate** to the sample (according to Instant claim 11);

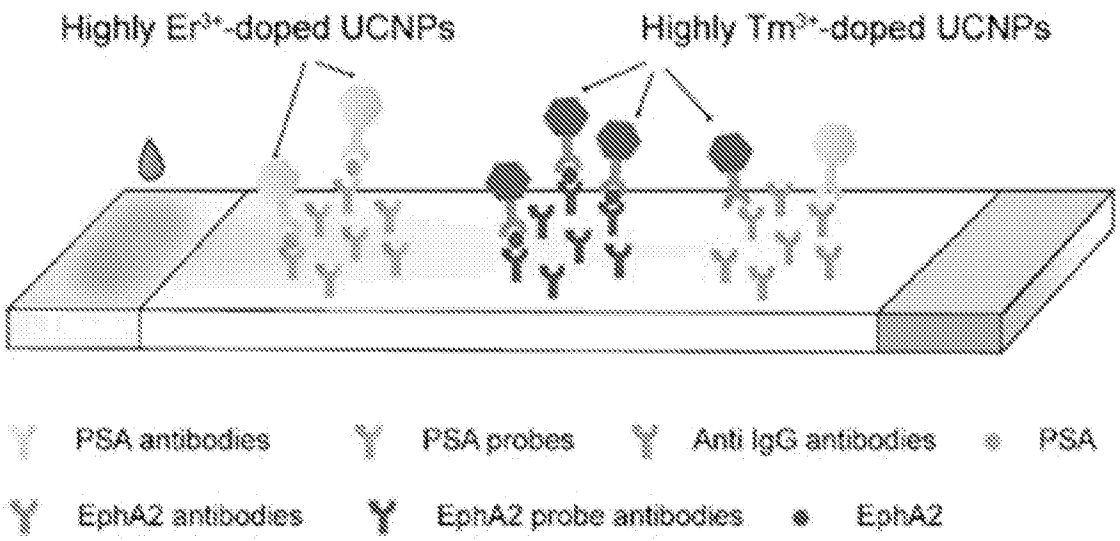
**an upconverting nanoparticle (UCNP) labeled conjugate in the sample** binds with the target analyte, if present; and detecting a label of the **upconverting nanoparticle (UCNP) labeled conjugate** at one or more of the at least two subwells to thereby determine whether the target analyte is present in the sample (according to Instant claim 1).

**He et al** teach throughout the document a platform and method for detecting target analytes from a fluid sample using upconverting nanoparticle (UCNP) based immunoassays on a lateral flow strip (LFS) (Fig. 1, and Abstract). UCNPs were used as labels for optical detection of the target analytes (pp. 12357, col. 1, para. 4, lines 1-4: In this study, we apply two kinds of UCNPs, highly doped by Er3+ ions to emit yellowish upconversion emissions and Tm3+ ions to emit purple upconversion emissions, as multicolor reporters in LFS.)

He et al further teach:

adding the upconverting nanoparticle (UCNP) labeled conjugate to the sample; and an upconverting nanoparticle (UCNP) labeled conjugate in the sample binds with the target analyte, if present; and detecting a label of the upconverting nanoparticle (UCNP) labeled conjugate at one or more of the at least two **test areas** to thereby determine whether the target analyte is present in the sample (Fig. 1D (as reproduced below) shows a lateral flow strip assay with two separate test areas for detection of PSA and EphA2 in a single sample. UCNP antibody conjugates are mixed with samples and applied to the sample pad area represented by the left side of the strip in Fig. 1D, allowed to bind with target analytes if present, then detected at the respective test areas for PSA and EphA2 (pp. S-7: UCNP reporters are transferred to a working buffer (pH6.8 MES buffer, 0.5% w/v tween 20, 1% w/v BSA). After mixing, sample solutions with UCNP reporters solutions are added to the sample pad of strip. After 10 mins, washing buffer (pH6.8 MES buffer, 0.5% w/v tween 20) is added to the sample pad. After 20 mins, the strip is detected by the reader.). Fig. 4 shows results of two-color PSA and EphA2 detection.).

**He et al, Fig. 1D**




It would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to modify the method for detecting target analytes from a sample using spatial multiplexed assays on microplates comprising subwells of Magliulo et al by substituting the use of peroxidase labeled conjugates (of Magliulo) with UCNP labeled conjugates added to the sample as disclosed by He et al (thereby also substituting the means for detection and optical analysis to ones equivalent to He et al), because the application of UCNP reporters in this way has high sensitivity in immunoassays (He et al, pp. 12356, col. 1, para. 1, lines 12-15: Using semiconductor quantum dots and ceramic upconversion nanoparticles (UCNPs), fluorescence-based LFS sensors have high sensitivity). Specifically, the modification would have been an obvious matter of a simple substitution of one known element (label) for another. In particular the prior art taught the base method of detecting a target as taught by Magliulo et al, which differed from that which is claimed only by the use of label. The prior art (Magliulo) also recognized at the time that either enzyme or fluorescent molecules were usable

as label for labeled conjugate. Even further, He et al recognized UCNP reports as fluorescent labels known to have high sensitivity. One having ordinary skill would have found it obvious to have used the fluorescent label of He et al, particularly because of the high sensitivity of UCNPs, and the results of the substitution would have been predictable (namely, sensitive detection would be achieved).

One having ordinary skill in the art would have had a reasonable expectation of success in combining the prior art references because Magliulo et al and He et al are similarly drawn to methods involving the optical detection of analytes on surfaces comprising regions coated with capture agents (He et al disclose test areas on a lateral flow strip, which are analogous to the subwells disclosed by Magliulo et al), and using labels that bind to the analytes to form conjugates. Furthermore, Magliulo et al disclose the use of fluorescent reporters as a known alternative to enzymatic reporters that would achieve similar predictable results for the purpose of immunoassays (Magliulo et al pp. 4933, col. 2, para. 3, line 5 – pp. 4934, col. 1, para. 1, line 3: Most of the commercially available immunoassays for bacteria detection are based on standard sandwich immunoassay (10), which involves the formation of an immunocomplex between an immobilized antibody, the target bacteria, and a secondary labeled antibody. Generally, the primary antibody is immobilized on beads (magnetic, silica, or gold particles), polystyrene microtiter plates, filter membranes, or directly on the surface of transducers, and the secondary antibody is labeled with enzymes or fluorescent molecules.).

Regarding claim 2, Magliulo et al further teach:

wherein the capture binding agent comprises at least a first capture binding agent and a second capture binding agent in different ones of the at least two subwells (Magliulo et al Fig. 1

shows the locations of immobilized monoclonal antibodies (MAbs) targeting different bacterial analytes in different subwells; and pp. 4934, col. 1, para. 6, lines 1-3: The developed method involves the immobilization of four MAbs, each specific for one bacterium, in the bottom of the different subwells of the main well.).

Regarding claim 3, Magliulo in view of He et al further teaches:

the UCNP labeled conjugate comprises a first UCNP labeled conjugate and a second UCNP labeled conjugate that are configured to bind to corresponding target analytes bound to the first and second capture binding agents, respectively, in the at least two subwells (He et al Fig. 1D and Fig. 4 show a two-color immunoassay with two types of UCNP labeled conjugates that specifically bind to their corresponding immobilized target analytes (i.e. PSA and EphA2) in different test areas. Magliulo et al Fig. 1 shows the locations of immobilized monoclonal antibodies (MAbs) targeting different target analytes in different subwells. Therefore, Magliulo et al in view of He et al teaches UCNP labeled conjugates configured to bind to corresponding targets in different subwells.).

Regarding claim 4, Magliulo et al in view of He et al further teaches:

wherein a UCNP of the first and second UCNP labeled conjugates is the same UCNP, and detecting a label of the labeled conjugate at one or more of the at least two subwells to thereby determine whether the target analyte is present in the sample comprises determining whether first and second target analytes are present based on which of the at least two subwells emit a signal from the UCNP labeled conjugate (Magliulo et al Fig. 1 shows a single peroxidase-label (i.e. HRP) conjugated to polyclonal antibodies (pp. 4934, col. 1, para. 6, lines 6-9:

Subsequently, a mixture of horseradish peroxidase (HRP)-labeled polyclonal antibodies was added and the peroxidase activity was determined by means of an enhanced luminol-based CL substrate.) and that the detection of analytes is based on the locations of chemiluminescence signals emitted from subwells loaded with known types of capture reagents. Therefore, Magliulo et al in view of He et al teaches the detection of analytes based on the subwell locations of signals emitted by a single type of UCNP used in UCNP labeled conjugates configured to bind to the analytes).

Regarding claim 5, Magliulo et al in view of He et al further teaches:

wherein detecting a UCNP label of the UCNP labeled conjugate at one or more of the at least two subwells comprises impinging photons on the at least two subwells such that the UCNP labeled conjugate emits a signal (He et al Fig. S4 and pp. S-8, lines 2-7: To read the emission signal of UCNPs reporters on strips, we use a purpose-built scanning confocal system, which is shown in Figure S4. A 976 nm laser is used to excite the UCNPs, with a home built power control unit including a half wave plate and a polarizer. The emission of UCNPs is collected by an objective lens (60X, NA 0.85 Edmund), then focused by a tube lens to an optical fibre, which is linked to a single photon avalanche detector (SPAD).; Therefore, Magliulo et al in view of He et al teaches the detection of UCNP labeled conjugates in subwells by impinging them with photons from a laser, resulting in the emission of a detectable signal).

Regarding claims 6 and 8, Magliulo et al in view of He et al further teaches:

determining a concentration of the target analyte based on a signal strength of a signal from the UCNP labeled conjugate, wherein the signal has a signal strength that corresponds to a

concentration of the target analyte, if present, in the sample; and wherein determining the concentration of the target analyte based on the signal strength of the signal from the UCNP labeled conjugate is based on an empirically-based model of actual experience (Magliulo Fig.1 shows an image containing intensity data for subwells representing test samples, control samples, and a calibration curve area; and pp. 4935, col. 2, para. 1, lines 15-19: The initial concentration of pathogens in the spiked samples was assessed by interpolating the measured CL signals on a calibration curve obtained from standard bacteria suspensions (concentrations ranging from 1 to 200 CFU/mL) that were subjected to the same enrichment procedure.; and pp. 4936, col. 1, para. 2, lines 15-20: Before data analysis, the blank signal was subtracted from all of the sample and calibration curve signals. Then, the initial concentration of each bacterium in the samples was determined by interpolation of the CL signals on the calibration curves obtained from the bacteria standard solutions.; Therefore, Magliulo et al in view of He et al teaches determining the concentration of analyte based on the strength of measured signals from UCNP conjugates bound to test capture binding agents (i.e. in the test subwells), and based on an empirically determined calibration curve of actual experience.).

Claims 7 and 10 are rejected under 35 U.S.C. 103 as being unpatentable over **Magliulo et al** (Magliulo, Maria, et al. "A rapid multiplexed chemiluminescent immunoassay for the detection of Escherichia coli O157: H7, Yersinia enterocolitica, Salmonella typhimurium, and Listeria monocytogenes pathogen bacteria." Journal of agricultural and food chemistry 55.13 (2007): 4933-4939.) as cited on the IDS dated 11/18/2022, in view of **He et al** (He, Hao, et al. "Quantitative lateral flow strip sensor using highly doped upconversion nanoparticles." Analytical chemistry 90.21 (2018): with Supplemental Information, 17 pages) as applied to

claims 1-6, and further in view of **Gong et al** (Gong, Yan, et al. "A portable and universal upconversion nanoparticle-based lateral flow assay platform for point-of-care testing." Talanta 201 (2019): 126-133.).

Regarding claims 7 and 10, **Magliulo et al in view of He et al** teaches the method of detecting a target analyte and/or performing a spatial multiplexed assay according to claims 1-6 as described above in detail.

Magliulo et al in view of He et al does not specifically teach:

wherein at least one of the first and second capture binding agents comprises a control capture binding agent and other of the first and second capture binding agents comprises a test capture binding agent, wherein a signal strength of a UCNP labeled conjugate bound to the test capture binding agent relative to a signal strength of a UCNP labeled conjugate bound to the control capture binding agent corresponds to the concentration of the target analyte, if present, in the sample (according to Instant claim 7);

and wherein the upconverting nanoparticle (UCNP) labeled conjugate is present on a bottom and/or side of the well prior to adding the sample to the well (according to Instant claim 10).

**Gong et al** teach throughout the document a platform and method for detecting target analytes from a fluid sample using upconverting nanoparticle (UCNP) based lateral flow strip immunoassays (Fig. 1, and Abstract).

Gong et al further teach:

wherein at least one of the first and second capture binding agents comprises a control capture binding agent and other of the first and second capture binding agents comprises a test capture binding agent (Fig. 1 shows embodiments of lateral flow assays with test and control lines comprising capture agents that respectively capture UCNP bound to analytes (i.e. in the test region) and unbound UCNPs (i.e. in the control region);

wherein a signal strength of a UCNP labeled conjugate bound to the test capture binding agent relative to a reference control corresponds to the concentration of the target analyte, if present, in the sample (pp. 128, col. 2, para. 1, lines 3-7: The photographs of the detection results were analyzed using our custom-designed App to calculate the optical density of test line (T) and background (B). The difference between "T" and "B" values is proportional to the concentration of the target analytes.);

and wherein the upconverting nanoparticle (UCNP) labeled conjugate is present on a surface prior to adding the sample to the surface (pp. 128, col. 1, para. 4, lines 4-11: The following processes were then performed before assembling the UCNP-based strip. First, the as-prepared UCNP probes were diluted by eluent buffer (pH 8.18, containing 0.85% [w/v] Tris, 1% [w/v] BSA and 10% [w/v] sucrose) before being dispensed on the conjugate pad at 40 μL/cm. Second, the control and test lines were fabricated by dispersing corresponding control probe and capture probe through Dispenser HM3030 (Goldbio Co., China) according to the sample to be tested.; Therefore, the UCNP labeled conjugate is added to the flow strip surface during the assembly process for the lateral flow strip, and is present prior to adding the sample).

It would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to modify the method for detecting target analytes from a sample using UCNP labeled conjugates in spatial multiplexed assays on microplates comprising subwells of Magliulo et al in view of He et al by combining the use of a test region and control region in the same assay (i.e. use of a test capture agent and a control capture agent in different fluidically connected subwells) along with analysis for determining the concentration of an analyte by comparing the test signal to the reference signal (i.e. comparing the test signal to the control signal to determine concentration) as disclosed by Gong et al, because this technique provides an internal reference that controls for assay variation, and by further incorporating the step of adding UCNP probes to the assay surface prior to adding the sample (i.e. adding UCNP conjugates to the subwells prior to adding the sample) as disclosed by Gong et al, because it simplifies the assay by eliminating the step of mixing UCNP probe with the sample (Gong et al).

One having ordinary skill in the art would have had a reasonable expectation of success in combining the prior art references, such to rely on controls/control areas as in Gong et al in the method of Magliulo et al, because they are similarly drawn to methods involving the optical detection of analytes on surfaces comprising regions coated with capture agents (Gong et al disclose test areas and control areas on a lateral flow strip, which are analogous to the subwells disclosed by Magliulo et al in view of He et al). Furthermore, Magliulo et al disclose controls and reference wells for comparing test results, and the modification would require altering the arrangement of capture agents already present on the microplate of Magliulo et al in view of He et al to include test and control capture agents in fluidically connected subwells, performing the corresponding analysis of measured signals, and adding UCNP conjugates to subwells using methods already disclosed by Magliulo et al prior to adding the sample.

Claims 9 is rejected under 35 U.S.C. 103 as being unpatentable over **Magliulo et al** (Magliulo, Maria, et al. "A rapid multiplexed chemiluminescent immunoassay for the detection of Escherichia coli O157: H7, Yersinia enterocolitica, Salmonella typhimurium, and Listeria monocytogenes pathogen bacteria." Journal of agricultural and food chemistry 55.13 (2007): 4933-4939.) as cited on the IDS dated 11/18/2022, in view of **He et al** (He, Hao, et al. "Quantitative lateral flow strip sensor using highly doped upconversion nanoparticles." Analytical chemistry 90.21 (2018): with Supplemental Information, 17 pages) as applied to claim 1, and further in view of **Kazakova et al** (Kazakova, Anna, et al. "Serological array-in-well multiplex assay reveals a high rate of respiratory virus infections and reinfections in young children." MSphere 4.5 (2019): 10-1128.) as cited on the IDS dated 11/18/2022.

Regarding claim 9, **Magliulo et al in view of He et al** teaches the method of detecting a target analyte and/or performing a spatial multiplexed assay according to claim 1 as described above in detail.

Magliulo et al in view of He et al does not specifically teach:

wherein the capture binding agent comprises a first half of a binding pair and the capture binding agent is bound to a surface of the at least two subwells via binding of the first half of the binding pair to a second half of the binding pair on the surface of the at least two subwells (according to Instant claim 9);

**Kazakova et al** teach throughout the document a multiplex microarray immunoassay (MAIA) that was developed for simultaneous detection of serum IgG antibodies against RSV, adenovirus, different influenza virus subtypes or vaccine, and several control antigens (Abstract and pp. 2, para. 3, lines 1-3). Kazakova et al teach an indirect detection configuration for detecting an analyte (i.e. human IgG), wherein the capture binding agent comprises a first half of a binding pair (i.e. a biotinylated antigen) and the capture binding agent is bound to a surface of a well of a microtiter plate via binding of the first half of the binding pair to a second half of the binding pair (i.e. streptavidin) on the surface of the well (Fig. 1 and pp. 2, para. 3, lines 3-5: Biotinylated antigens were printed in a 4 × 5 format on the bottom of streptavidin-coated 96-well microtiter plate wells.).

It would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to modify the method for detecting target analytes from a sample using UCNP labeled conjugates in spatial multiplexed assays on microplates comprising subwells of Magliulo et al in view of He et al by substituting the capture agent with a binding pair that acts as a capture binding agent (i.e. a biotinylated capture reagent that is bound to a streptavidin that is bound to the subwells surfaces) as disclosed by Kazakova et al, because this method is desirable for depositing a variety of capture reagents targeting various analytes using a standard printing protocol optimized for the biotin/streptavidin binding reaction (Kazakova et al, pp. 12, para. 6, lines 2-3: Biotinylated antigens were printed to streptavidin-coated wells into an array format using a noncontact microarrayer Nanoplotter 2.1 (GeSiM, Germany) with 70% humidity, 80-V voltage, 50-μs pulse, 100-Hz frequency, and 2 droplets/spot.). This constitutes a simple substitution of a known method for binding the capture agent directly to a surface with

another known method of indirectly binding the capture agent to obtain the same predictable result of immobilizing the capture agent to the surface of the subwell in order to capture an analyte.

One having ordinary skill in the art would have had a reasonable expectation of success in combining the prior art references because they are similarly drawn to methods involving the detection of analytes on microplates comprising regions coated with capture agents (Kazakova et al teach assays using microtiter plates, which are analogous to the assays on subwells disclosed by Magliulo et al in view of He et al). Magliulo et al disclose structure necessary for the modifications, including a method for the immobilization of proteins (i.e. monoclonal antibodies) which would be suitable for binding streptavidin to the subwells, for use as capture agents. Magliulo et al also disclose the concept of indirect capture through the binding of secondary antibodies for detection purposes. Furthermore, protocols required for the biotinylation of capture reagents of interest were well known in the art, as disclosed by Kazakova et al (pp.12, para. 4).

Claim 12 is rejected under 35 U.S.C. 103 as being unpatentable over **Magliulo et al** (Magliulo, Maria, et al. "A rapid multiplexed chemiluminescent immunoassay for the detection of Escherichia coli O157: H7, Yersinia enterocolitica, Salmonella typhimurium, and Listeria monocytogenes pathogen bacteria." Journal of agricultural and food chemistry 55.13 (2007): 4933-4939.) as cited on the IDS dated 11/18/2022, in view of **He et al** (He, Hao, et al. "Quantitative lateral flow strip sensor using highly doped upconversion nanoparticles." Analytical chemistry 90.21 (2018): with Supplemental Information, 17 pages) as applied to claim 1, and further in view of **HCG ELISA Instructions** (DRG Instruments GmbH, HCG ELISA

EIA-1469, Version 10.0, Dated: 12/2020, 36 pages. Downloaded from: https://www.drg-diagnostics.de/files/eia-1469_ifu--hcg_2020-12-17_endeitesfr.pdf).

Regarding claim 12, **Magliulo et al in view of He et al** teaches the method of detecting a target analyte and/or performing a spatial multiplexed assay according to claim 1 as described above in detail.

Magliulo et al in view of He et al does not specifically teach:

wherein the target analyte is human chorionic gonadotropin (hCG).

**HCG ELISA Instructions** teaches throughout the document a solid phase enzyme-linked immunosorbent assay (ELISA) wherein the target analyte is human chorionic gonadotropin (hCG) (see pp. 2, INTENDED USE). The microtiter wells are coated with a monoclonal [mouse] antibody directed towards a unique antigenic site on an hCG molecule. An aliquot of patient sample containing endogenous hCG is incubated in the coated well with enzyme conjugate, which is a monoclonal antibody directed against the alpha-chain of hCG conjugated with horseradish peroxidase. After incubation the unbound conjugate is washed off. The amount of bound peroxidase is proportional to the concentration of hCG in the sample. Having added the substrate solution, the intensity of colour developed is proportional to the concentration of hCG in the patient sample. (see pp. 2, PRINCIPLE OF THE TEST)

It would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to modify the method for detecting target analytes from a sample using UCNP labeled conjugates in spatial multiplexed assays on microplates comprising subwells of Magliulo et al in view of He et al by incorporating the use of human chorionic gonadotropin as a target analyte (i.e. by using an appropriate capture agent and a labeled conjugate that binds to hCG) as disclosed by HCG ELISA Instructions, because "HCG Assays are used for the early detection of pregnancy" (HCG ELISA Instructions, pp. 2, Summary and Explanation).

One having ordinary skill in the art would have had a reasonable expectation of success in combining the prior art references because they are similarly drawn to methods involving the optical detection of analytes on substrates comprising regions coated with capture agents (HCG ELISA Instructions discloses an assay on a microtiter plate, which is analogous to the assays on subwells disclosed by Magliulo et al in view of He et al). Magliulo et al disclose structure necessary for the modifications, including a method for the immobilization of monoclonal antibodies in subwells for use as capture agents, and the binding of secondary antibodies for detection purposes. Furthermore, Magliulo et al disclose capture agents and detection antibodies targeting various analytes, and the modification would simply require substituting them with capture agents and UCNP labeled detection antibodies targeting hCG.

Claim 13 is rejected under 35 U.S.C. 103 as being unpatentable over **Magliulo et al** (Magliulo, Maria, et al. "A rapid multiplexed chemiluminescent immunoassay for the detection of Escherichia coli O157: H7, Yersinia enterocolitica, Salmonella typhimurium, and Listeria monocytogenes pathogen bacteria." Journal of agricultural and food chemistry 55.13 (2007):

4933-4939.) as cited on the IDS dated 11/18/2022, in view of **He et al** (He, Hao, et al. "Quantitative lateral flow strip sensor using highly doped upconversion nanoparticles." Analytical chemistry 90.21 (2018): with Supplemental Information, 17 pages) as applied to claim 1, and further in view of **PLGF ELISA Instructions** (DRG Instruments GmbH, PLGF ELISA EIA-4529, Version 13.0, Dated: 07/2017, 19 pages. Downloaded from: https://www.drg-diagnostics.de/files/eia-4529_ifu--plgf_2017-07-11_ende.pdf).

Regarding claim 13, **Magliulo et al in view of He et al** teaches the method of detecting a target analyte and/or performing a spatial multiplexed assay according to claim 1 as described above in detail.

Magliulo et al in view of He et al does not specifically teach:

wherein the target analyte is placental growth factor (PlGF).

**PLGF ELISA Instructions** teaches throughout the document a solid phase enzyme-linked immunosorbent assay (ELISA) wherein the target analyte is placenta growth factor (PLGF) (see pp. 2, INTENDED USE). The microtiter wells are coated with a monoclonal antibody directed towards a unique antigenic site of the PLGF molecule. An aliquot of patient sample containing endogenous PLGF is incubated in the coated well. After a washing step a biotin-linked polyclonal antibody specific for PLGF is added to the wells. Following a wash to remove any unbound antibody a streptavidin HRP enzyme complex is added to the wells. After incubation the unbound enzyme complex is washed off. The amount of bound peroxidase is

proportional to the concentration of PLGF in the sample. Having added the substrate solution, the intensity of colour developed is proportional to the concentration of PLGF in the patient sample. (see pp. 2, PRINCIPLE OF THE TEST)

It would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to modify the method for detecting target analytes from a sample using UCNP labeled conjugates in spatial multiplexed assays on microplates comprising subwells of Magliulo et al in view of He et al by incorporating the use of placental growth factor as a target analyte (i.e. by using an appropriate capture agent and a labeled conjugate that binds to PLGF) as disclosed by PLGF ELISA Instructions, because the PLGF assay "can be used as a diagnostic aid to estimate the probability of preeclampsia for pregnant women" (PLGF ELISA Instructions, pp. 2, INTENDED USE).

One having ordinary skill in the art would have had a reasonable expectation of success in combining the prior art references because they are similarly drawn to methods involving the optical detection of analytes on substrates comprising regions coated with capture agents (PLGF ELISA Instructions discloses an assay on a microtiter plate, which is analogous to the assays on subwells disclosed by Magliulo et al in view of He et al). Magliulo et al disclose structure necessary for the modifications, including a method for the immobilization of monoclonal antibodies in subwells for use as capture agents, and the binding of secondary antibodies for detection purposes. Furthermore, Magliulo et al disclose capture agents and detection antibodies targeting various analytes, and the modification would simply require substituting them with capture agents and UCNP labeled detection antibodies targeting PLGF.

## *Conclusion*

No claims are allowed.

## *Examiner's Contact Information*

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Ramdane Harouaka whose telephone number is (571) 270-0523. The examiner can normally be reached Monday-Friday 8am-5pm ET.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Bao-Thuy Nguyen can be reached on (571) 272-0824. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/RAMDANE HAROUAKA/
Examiner, Art Unit 1677

/ELLEN J MARCSISIN/
Primary Examiner, Art Unit 1677